Synthetic BOAMP Linkage Algorithm Benchmark Report

Generated from the corrected synthetic benchmark notebook

2026-07-30

Technical Summary

Gradient boosting is the strongest current linker on the corrected synthetic benchmark. It reaches precision **0.729**, end-to-end recall **0.270**, and end-to-end pair-F1 **0.394** across the held-out evaluation grid.

The result is not mainly a classifier-only story. It is a pipeline story: the production candidate generator exposes only **10,825** of **31,426** in-scope truth links to scoring, a weighted blocking recall of **34.4%** and an unweighted world-average recall of **0.323**. This is why gradient boosting has fixed-candidate F1 **0.756** but end-to-end F1 only **0.394**.

Logistic regression recovers essentially the same number of true links as gradient boosting (8,496 versus 8,494), but accepts far more links (21,878 versus 11,648). The extra volume lowers precision from **0.729** to **0.388**. The current weighted composite and Fellegi-Sunter-style baseline remain useful comparators, but they are not the preferred scorers under this benchmark.

Gradient Boosting Wins, Mostly By Reducing False Positives

The four algorithms were evaluated on the same held-out candidate-pair universe. Gradient boosting accepts fewer links than logistic regression and the composite baseline, while retaining nearly the same true-positive count as logistic regression. That makes it the best current choice for synthetic benchmark ranking and the least risky of the tested options for downstream survival-analysis pilots.

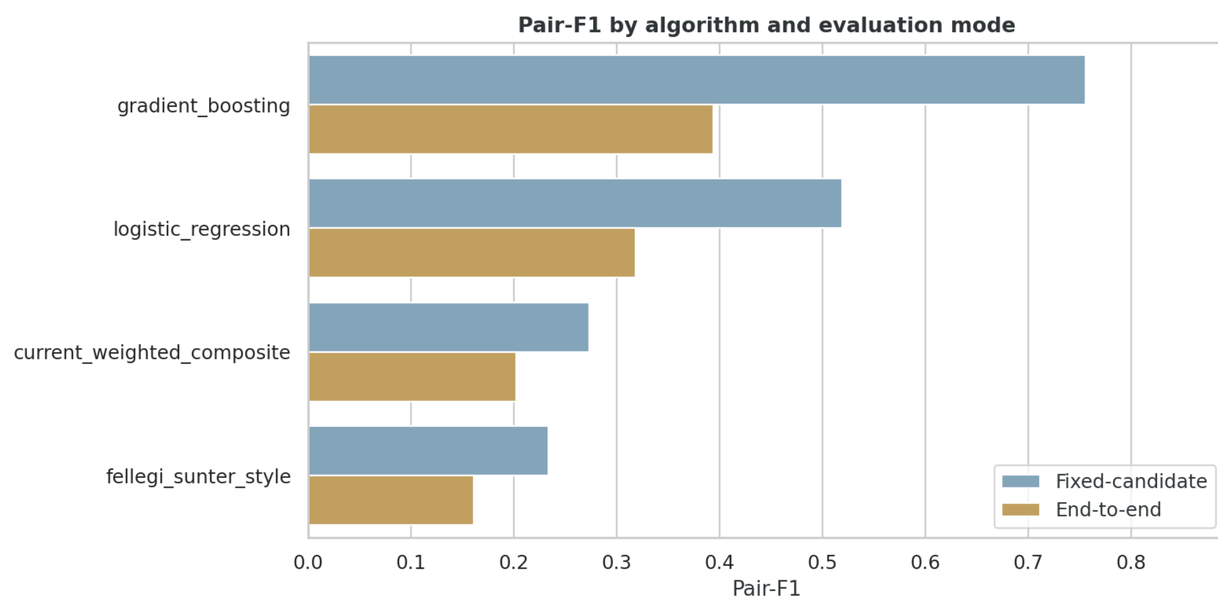

Figure 1: Pair-F1 by algorithm and evaluation mode. Fixed-candidate performance is much higher because unreachable truth links are excluded from that denominator.

Table 1: Aggregate held-out performance by algorithm.

Algorithm	Predicted links	True positives	Precision	Fixed-candidate recall	End-to-end recall	End-to-end F1
Gradient boosting	11,648	8,494	0.729	0.785	0.270	0.394
Logistic regression	21,878	8,496	0.388	0.785	0.270	0.319
Current weighted composite	48,072	8,061	0.168	0.745	0.257	0.203
Fellegi-Sunter style	34,581	5,308	0.153	0.490	0.169	0.161

Compared with the current weighted composite, gradient boosting improves precision by **4.3x** and end-to-end F1 by **1.9x**.

Candidate Generation Is The Binding Constraint

End-to-end performance is much lower than fixed-candidate performance because most truth links never enter the candidate set. The evaluation run scored **767,431** candidate pairs over **31,426** in-scope truth links, but only **10,825** truth links were reachable by the production candidate generator.

Table 2: Candidate-generation reachability by evaluation scenario.

Scenario	Worlds	Candidate pairs	Truth links	Reachable truth	Mean blocking recall	Min	Max
central_provisional	4	68,528	6,635	2,586	0.394	0.341	0.427
difficult	4	140,698	3,999	1,039	0.262	0.214	0.284
easier	4	103,267	10,127	3,966	0.397	0.342	0.424
moderate	4	68,528	6,635	2,586	0.394	0.341	0.427
stress	4	386,410	4,030	648	0.166	0.132	0.204

The stress scenario makes this clearest: it creates 386,410 candidate pairs but exposes only 648 of 4,030 truth links to scoring. No scoring model can recover links that were never generated. Therefore the next major gain should come from candidate generation, not only from another classifier.

Scenario Results Show Robust Ranking But Different Difficulty

Gradient boosting ranks first in every evaluated scenario. The ranking is stable, but absolute performance changes sharply as the benchmark gets harder.

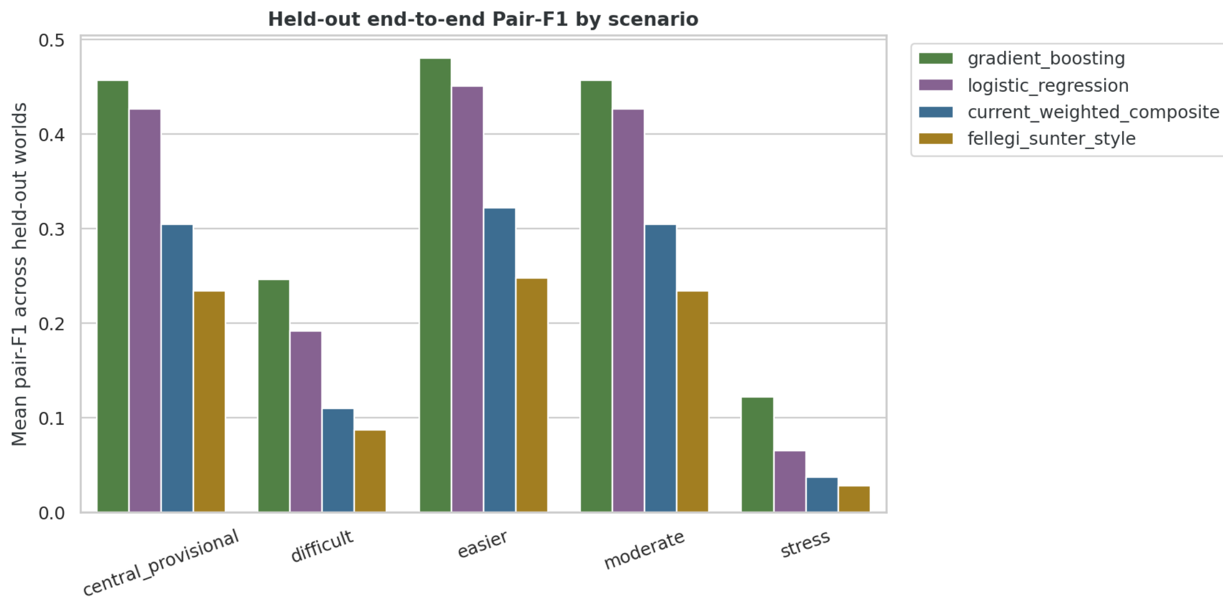

Figure 2: Held-out end-to-end pair-F1 by scenario. Gradient boosting leads throughout, but stress and difficult scenarios lower all algorithms.

Table 3: Mean held-out performance by scenario and algorithm.

Scenario	Algorithm	Mean F1	Mean precision	Mean recall	SD F1
central_provisional	Gradient boosting	0.457	0.800	0.321	0.033
central_provisional	Logistic regression	0.427	0.646	0.319	0.038
central_provisional	Current weighted composite	0.305	0.309	0.303	0.050
central_provisional	Fellegi-Sunter style	0.235	0.284	0.202	0.043
difficult	Gradient boosting	0.246	0.493	0.165	0.026
difficult	Logistic regression	0.192	0.212	0.176	0.037
difficult	Current weighted composite	0.110	0.085	0.160	0.031
difficult	Fellegi-Sunter style	0.087	0.082	0.098	0.027
easier	Gradient boosting	0.481	0.861	0.334	0.038
easier	Logistic regression	0.451	0.727	0.328	0.038
easier	Current weighted composite	0.322	0.334	0.313	0.051
easier	Fellegi-Sunter style	0.248	0.307	0.209	0.040
moderate	Gradient boosting	0.457	0.800	0.321	0.033
moderate	Logistic regression	0.427	0.646	0.319	0.038
moderate	Current weighted composite	0.305	0.309	0.303	0.050
moderate	Fellegi-Sunter style	0.235	0.284	0.202	0.043
stress	Gradient boosting	0.122	0.260	0.080	0.028
stress	Logistic regression	0.065	0.052	0.090	0.020
stress	Current weighted composite	0.037	0.024	0.089	0.013
stress	Fellegi-Sunter style	0.028	0.020	0.051	0.010

The easier, central, and moderate settings support useful discrimination among algorithms. The difficult and stress settings are more diagnostic of failure modes. In stress, gradient boosting still leads, but end-to-end F1 falls to **0.122** because blocking recall and candidate ambiguity both worsen. Central and moderate are identical in this executed run, so they should not be interpreted as separate difficulty levels until their scenario definitions diverge.

Mechanical Bias Remains Visible

The benchmark shows systematic performance differences by identifier quality, candidate count, schema, and CPV availability. These are expected linkage failure modes, but they matter because survival analysis could inherit them.

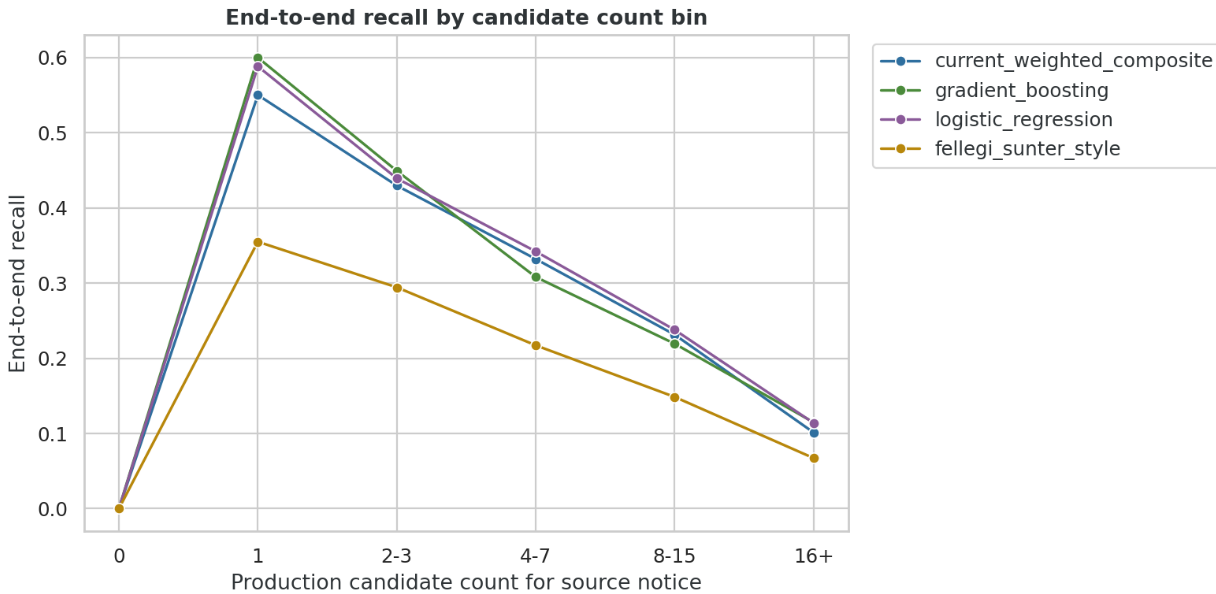

Figure 3: End-to-end recall by candidate count bin. Crowded candidate lists sharply reduce recall.

For gradient boosting, end-to-end recall is **0.600** when a source has exactly one candidate, but only **0.115** when it has sixteen or more candidates. This is the strongest bias diagnostic: crowded candidate environments are under-linked.

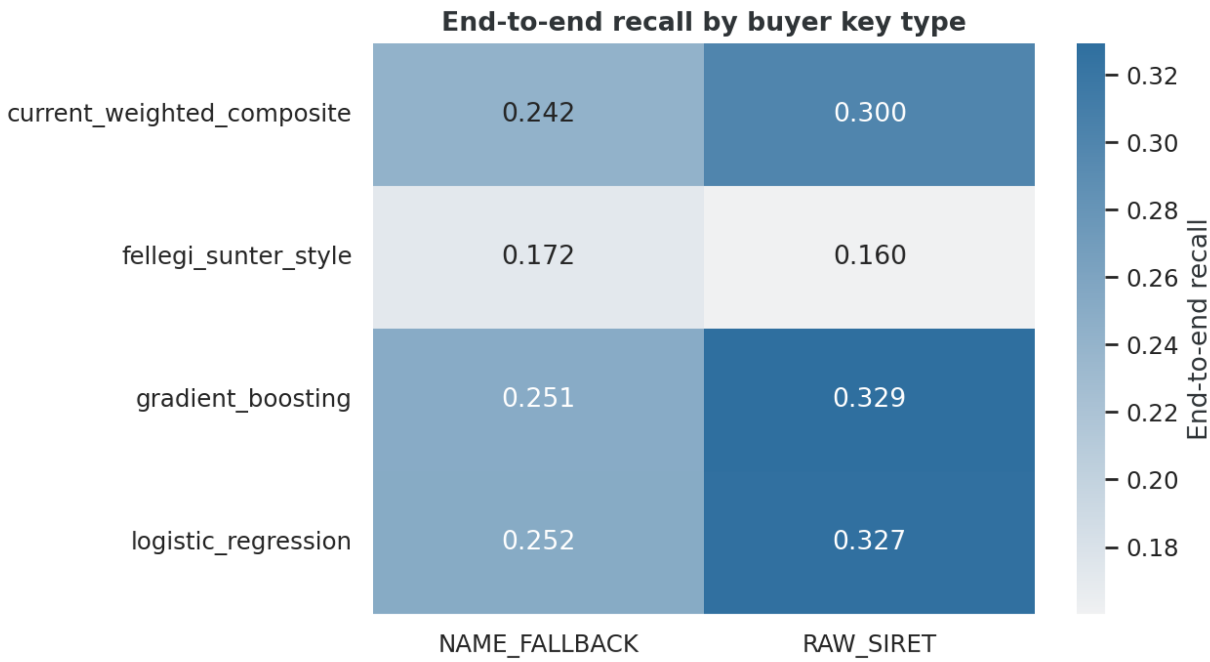

Figure 4: End-to-end recall by buyer key type. Cleaner identifiers remain easier to link.

Identifier quality also matters. Gradient boosting recall is **0.329** for RAW_SIRET truth links versus **0.251** for NAME_FALLBACK. CPV availability matters as well: recall is **0.286** when CPV is present and **0.133** when CPV is missing.

Thresholds And Model Setup

The current composite and Fellegi-Sunter thresholds are frozen. Logistic regression and gradient boosting thresholds are selected on central calibration worlds 005 and 006 by F1 maximization. Training uses central worlds 001 through 004; evaluation uses worlds 007 through 010 across central provisional, easier, moderate, difficult, and stress scenarios.

Table 4: Algorithm score columns and thresholds used in the executed benchmark.

Algorithm	Score column	Threshold	Threshold source
Current weighted composite	composite_score	0.343167	production primary threshold
Fellegi-Sunter style	fs_log_odds	0	frozen zero log-odds threshold
Logistic regression	logistic_score	0.824308	calibration F1 maximum
Gradient boosting	gradient_boosting_score	0.265103	calibration F1 maximum

The supervised models use pair-level scoring features already produced by the project pipeline: text similarity, CPV similarity, time score, buyer reliability score, gap features, candidate count, schema family, buyer key type, CPV missingness, and duration missingness indicators. The report does not claim these features are unbiased; the bias diagnostics above show that several are mechanically important.

Limitations, Robustness, And Interpretation

These results are valid for the current synthetic benchmark, not for real BOAMP ground truth. The synthetic truth labels are generated, so model ranking can still contain generator-mechanism bias. The report supports the claim that gradient boosting is best among the four tested linkers on this synthetic benchmark; it does not prove real-world BOAMP linkage accuracy.

The aggregate metrics are corrected to use a global pair key containing scenario, world, and notice pair. This avoids collisions from repeated synthetic notice IDs across generated worlds. The notebook was rerun after that correction and all tables in this report use the corrected outputs.

The fixed-candidate view is useful for comparing scorers, but the end-to-end view is the decision-relevant pipeline metric. If downstream survival analysis needs low false positives, gradient boosting is the best candidate among the tested models. If it needs high recall, the current candidate generation must be widened or redesigned.

Recommended Next Steps

1. Use gradient boosting as the primary supervised linker for the next synthetic benchmark round.
2. Keep logistic regression as the interpretable supervised baseline, and keep the current composite and Fellegi-Sunter style models as frozen reference baselines.
3. Improve candidate generation before treating linkage as production-ready. Test wider temporal windows, alternate buyer-name blocking, CPV fallback blocking, and high-activity buyer handling.
4. Add a manually labeled real BOAMP validation set before making claims about real BOAMP accuracy.
5. Run survival-analysis sensitivity with at least three linkage layers: current composite, gradient boosting, and a high-precision conservative subset.

Further Questions

1. How much end-to-end recall is recovered by widening candidate generation without creating too many false-positive candidates?
2. Are high-activity buyers under-linked enough to bias survival estimates?
3. Can a conservative gradient-boosting threshold provide a high-precision survival layer while a broader threshold supports recall sensitivity?
4. Do manually labeled real BOAMP pairs preserve the same algorithm ranking observed in synthetic truth?

Source Artifacts

- Benchmark version: `v0_3_temporal_candidate_revision`
- Executed notebook: `notebooks/14_linkage_algorithm_benchmark.ipynb`
- Summary table: `reports/tables/synthetic_benchmark/v0_3_temporal_candidate_revision/linkage_algorithm_benchmark/summary_metrics.csv`
- Scenario table: `reports/tables/synthetic_benchmark/v0_3_temporal_candidate_revision/linkage_algorithm_benchmark/scenario_summary_metrics.csv`
- Bias diagnostics: `reports/tables/synthetic_benchmark/v0_3_temporal_candidate_revision/linkage_algorithm_benchmark/recall_slices.csv`